

E-commerce: Customer & Sales Behavior

Project Overview

Dataset: Online Retail dataset

Scenario: A mid-size online store wants to understand who its best customers are and which products drive revenue

Questions:

1. Who are the top 10 customers by total spend?
2. What % of revenue comes from the top 20% of customers (Pareto check)?
3. What are the top 5 best-selling products by quantity and by revenue?
4. Which month had the highest sales, and what likely drove it?
5. What are the top 10 Country/Region by Total Orders and Revenue?
6. Which products are frequently bought?

Deliverable: Power BI dashboard with revenue trend, top customers, top products, and a "VIP customer" recommendation.

Step 1 — Excel

The original dataset contained **1,048,575** transaction records. During the data cleaning process, **262,975** records with missing Customer IDs, **18,160** return/cancellation transactions (negative quantities), and **70** zero-priced transactions were removed. The final analytical dataset contains **767,370** valid sales transactions, which was used for all subsequent SQL analysis and Power BI dashboard development.

Cleaning Step	Records Removed
Original Records	10,48,575
Missing Customer ID	2,62,975
Returns/Cancellations Moved	18,160
Price = 0 Removed	70
Duplicate Records Removed	34,150
Final Clean Dataset	7,67,370

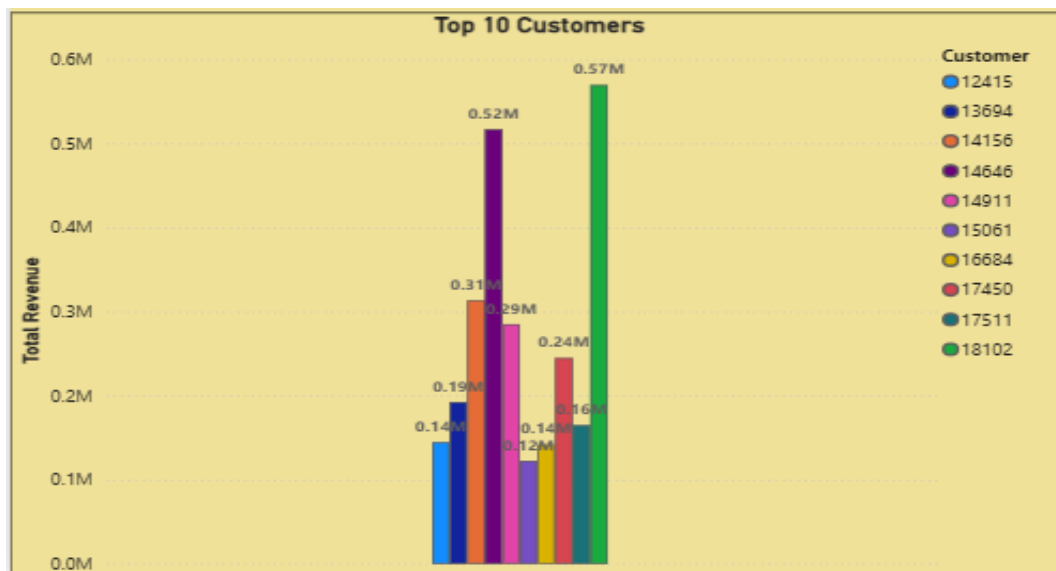
— SQL

Ran 6 queries to answer: who are the top customers, how concentrated is revenue (Pareto), what are the top products, what's the monthly trend, how does average order value vary by country, and what products are commonly bought together.

Key Findings

1. Who are the top 10 customers by total spend?

	CustomerID	total_spend
▶	18102	569501.50
	14646	516874.50
	14156	313437.62
	14911	285118.84
	17450	244784.25
	13694	192509.53
	17511	164753.55
	12415	144458.37
	16684	141740.79
	15061	122493.16



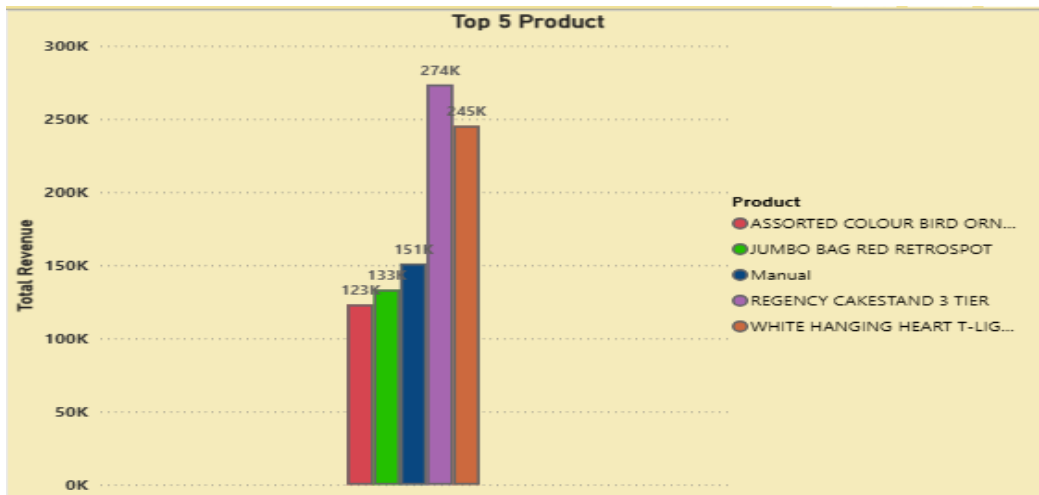
2. What % of revenue comes from the top 20% of customers (Pareto check)?

pct_revenue_from_top20_customers

77.01

3. What are the top 5 best-selling products by quantity and by revenue?

StockCode	Description	Total_Revenue	Total_Quantity
22423	REGENCY CAKESTAND 3 TIER	273694.42	23780
85123A	WHITE HANGING HEART T-LIGHT HOLDER	245488.77	91173
M	Manual	151350.21	8574
85099B	JUMBO BAG RED RETROSPOT	133467.00	73806
84879	ASSORTED COLOUR BIRD ORNAMENT	123330.16	77584



4. Which month had the highest sales, and what likely drove it?

Order Month format is (Year-Month)

Order_Month	Monthly_Revenue
09-12	683504.01
10-01	555802.67
10-02	504558.95
10-03	696978.47
10-04	591982.00
10-05	597833.38
10-06	636371.13
10-07	589736.17
10-08	602224.60
10-09	829013.95
10-10	1033112.01
10-11	1166460.02
10-12	570422.73
11-01	568101.31
11-02	446084.92
11-03	594081.76
11-04	468374.33

11-05	677355.15
11-06	660046.05
11-07	598962.90
11-08	644051.04
11-09	950690.20
11-10	1035642.45
11-11	1156205.61
11-12	99305.12

5. What are the top 10 Country/Region by Total Orders and Revenue?

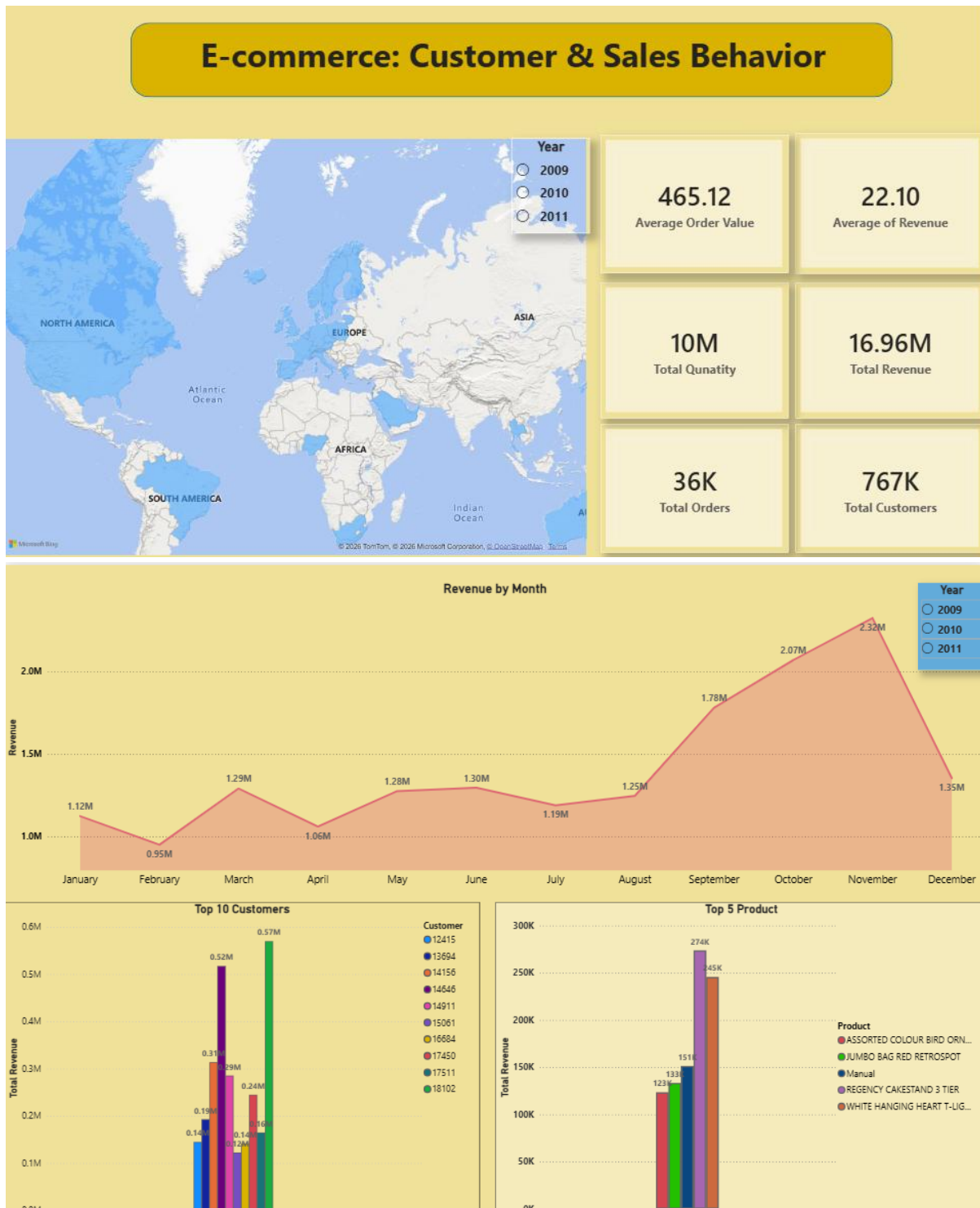
Country	Total_Orders	Total_Revenue
United Kingdom	33085	14009957.77
EIRE	560	610268.57
Netherlands	225	542310.07
Germany	775	417798.41
France	601	343984.68
Australia	95	169283.46
Spain	152	108016.28
Switzerland	90	100061.94
Sweden	103	91277.82
Denmark	42	68411.79

6. Which products are frequently bought?

Product	Product_Name	Times_Bought
85123A	WHITE HANGING HEART T-LIGHT HOLDER	4982
22423	REGENCY CAKESTAND 3 TIER	3302
84879	ASSORTED COLOUR BIRD ORNAMENT	2666
85099B	JUMBO BAG RED RETROSPOT	2611
47566	PARTY BUNTING	2087
20727	LUNCH BAG BLACK SKULL.	2023
21034	REX CASH+CARRY JUMBO SHOPPER	1923
22382	LUNCH BAG SPACEBOY DESIGN	1873
21232	STRAWBERRY CERAMIC TRINKET BOX	1859
21754	HOME BUILDING BLOCK WORD	1844

— KPI and Dashboard

: Power BI dashboard with revenue trend, top customers, top products, and a "VIP customer" recommendation.



Business Recommendations

- Develop a **VIP loyalty program** focused on the highest-value 20% of customers to improve retention and maximize customer lifetime value.
 - Review pricing and profitability for high-volume but comparatively low-revenue products to optimize pricing strategy or intentionally position them as traffic-driving products.
 - Increase inventory levels and marketing investment ahead of the identified peak sales months to prevent stock shortages and capitalize on seasonal demand.
 - Launch **bundle offers and cross-selling campaigns** using the frequently purchased product combinations to increase average order value.
 - Reduce the number of missing Customer IDs by encouraging account creation or loyalty program registration during checkout, improving the quality of future customer analytics.
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Tools Used

- **Microsoft Excel** – Data Cleaning & Preparation
- **MySQL** – Business Analysis & SQL Queries
- **Power BI** – Dashboard Development & Business Intelligence